CASE FILE: OPERATION GHOST KEY

* **Case Metadata:**
    * **Operation Name:** OPERATION: GHOST KEY
    * **Incident ID:** INC-2026-0826-HSM
    * **Facility:** Vanguard Cybernetics R&D Facility, Sector 7 (Sub-Levels 1 to 4)
    * **Incident Timestamp:** 2026-08-26T22:47:00Z
    * **Target Asset:** HSM Unit 09-Alpha (Hardware Security Module, Enterprise Cryptographic Key Vault)
    * **Security Clearance Level:** Level 5, Black Baseline
    * **Difficulty Rating:** 5-Star Extreme

* **Investigation Parameters:**
    * **Confidence Threshold:** 0.85
    * **Database Integrity Status:** Compromised at 23:15
    * **Reliability Weights:**
        * Device Telemetry: 0.95
        * Biometric Logs: 0.88
        * CCTV Feed: 0.85
        * RFID Access Logs: 0.70
        * Witness Statement: 0.55
        * Live System Alerts: 0.45
        * Anonymous Tip: 0.30

SUSPECTS

* **Dr. Aris Thorne (SUSPECT A):**
    * **Role:** Lead Cryptographic Engineer
    * **Badge ID:** EMP-8842
    * **Statement:** "I completed my shift at 19:30, handed over the calibration log, and left the building at 20:00. I was at the Downtown Bistro by 20:30 and stayed there until midnight. I never authorized any HSM extraction."
    * **Secret Motive:** Illicit IP extraction of low-level cipher specs for an external buyer, unrelated to the physical HSM unit.

* **Elena Rostova (SUSPECT B):**
    * **Role:** Senior Infrastructure and Cyber Defense Officer
    * **Badge ID:** EMP-3109
    * **Statement:** "I never entered the restricted R&D floor after 18:00. When the alarm sounded at 22:47, I was working remotely from home via encrypted VPN."
    * **Secret Motive:** Initiated an emergency offline extraction protocol to preserve HSM keys from an active, undetectable zero-day exploit.

* **Marcus Vance (SUSPECT C):**
    * **Role:** Facilities and Physical Security Operations Chief
    * **Badge ID:** EMP-1052
    * **Statement:** "I performed a standard sweep of Sector 7 at 21:00. All doors were secured, and the HSM vault intact. I did not touch the access control database."

    * **Secret Motive:** Hiding physical maintenance breaches and accepting bribes to allow an unauthorized rogue hardware tap on the facility spine.

* **Kaelen Voss (SUSPECT D):**
    * **Role:** Lead System Architect and Database Administrator
    * **Badge ID:** EMP-4401
    * **Statement:** "I was performing scheduled system maintenance on the main database server from 20:00 to 22:30 in server room B. Any log discrepancies are due to a scheduled epoch syncing routine."
    * **Secret Motive:** Exploited database access to alter access logs and sell badge-cloning software to corporate espionage groups.

EVIDENCE ITEMS

* **EVI 001:**
    * **Source Type:** RFID Access Logs
    * **Timestamp:** 2026-08-26T19:42:00Z
    * **Description:** RFID Badge EMP-8842 (Dr. Aris Thorne) scanned at Sector 7 R&D floor main turnstile.
    * **Status:** Altered
    * **Reliability Score:** 0.70
    * **Cross Reference:** EVI 004
    * **Anomaly Note:** Log entry insertion time differs from network epoch timestamp by 142 seconds.

* **EVI 002:**
    * **Source Type:** CCTV Feed
    * **Timestamp:** 2026-08-26T19:51:00Z
    * **Description:** Camera CAM-7B captures an individual wearing Dr. Thorne's coat and lab badge entering the Lab 3 door keyway.
    * **Status:** Altered
    * **Reliability Score:** 0.85
    * **Cross Reference:** EVI 005
    * **Anomaly Note:** Facial height to shoulder ratio does not match Dr. Thorne's biometric profile.

* **EVI 003:**
    * **Source Type:** RFID Access Logs
    * **Timestamp:** 2026-08-26T19:54:00Z
    * **Description:** Vault 9-Alpha opened using Master Override Code 55-Delta, assigned to Marcus Vance.
    * **Status:** Genuine
    * **Reliability Score:** 0.70
    * **Cross Reference:** EVI 008
    * **Anomaly Note:** Physical vault latch sensors trigger a physical removal flag.

* **EVI 004:**
    * **Source Type:** Device Telemetry
    * **Timestamp:** 2026-08-26T19:42:00Z
    * **Description:** Dr. Aris Thorne's authenticated mobile smartphone registered on Metro Tower 4 Cell Tower, 3.2 miles from the facility.
    * **Status:** Genuine

    * **Reliability Score:** 0.95
    * **Cross Reference:** EVI 001
    * **Anomaly Note:** Proves physical badge EMP-8842 was cloned or used by a proxy.

* **EVI 005:**
    * **Source Type:** CCTV Feed
    * **Timestamp:** 2026-08-26T19:57:00Z
    * **Description:** Camera CAM-2A captures Dr. Thorne's badge holder purchasing coffee at the facility cafeteria (Ground Floor).
    * **Status:** Completely Misleading
    * **Reliability Score:** 0.85
    * **Cross Reference:** EVI 002
    * **Anomaly Note:** Physically impossible to travel from Lab 3 (Sub-level 2) to the Cafeteria (Ground Floor) in 3 minutes.

* **EVI 006:**
    * **Source Type:** Biometric Logs
    * **Timestamp:** 2026-08-26T19:53:12Z
    * **Description:** Physical retinal scanner at the Lab 3 rear emergency airlock logged a scan from Elena Rostova.
    * **Status:** Genuine
    * **Reliability Score:** 0.88
    * **Cross Reference:** EVI 009
    * **Anomaly Note:** Directly contradicts Elena Rostova's statement of never entering the restricted R&D floor after 18:00.

* **EVI 007:**
    * **Source Type:** Device Telemetry
    * **Timestamp:** 2026-08-26T22:30:00Z
    * **Description:** Internal IDS alert reporting a foreign payload attempting a memory dump of HSM 09-Alpha via a zero-day SSH vulnerability.
    * **Status:** Genuine
    * **Reliability Score:** 0.95
    * **Cross Reference:** EVI 010
    * **Anomaly Note:** Source IP traced back to Marcus Vance's compromised internal maintenance tap.

* **EVI 008:**
    * **Source Type:** Anonymous Tip
    * **Timestamp:** 2026-08-26T21:15:00Z
    * **Description:** Encrypted message sent to the security hotline stating that Thorne sold the cryptographic root keys to foreign buyers at 20:00.
    * **Status:** Completely Misleading
    * **Reliability Score:** 0.30
    * **Cross Reference:** EVI 004
    * **Anomaly Note:** Fabricated tip sent to frame Dr. Thorne and divert attention from the active breach.

* **EVI 009:**
    * **Source Type:** Device Telemetry
    * **Timestamp:** 2026-08-26T19:55:00Z

    * **Description:** Air gapped safe deposit enclosure activated in Elena Rostova's secure subterranean workspace (B-4 Vault).
    * **Status:** Genuine
    * **Reliability Score:** 0.95
    * **Cross Reference:** EVI 006
    * **Anomaly Note:** Confirms physical relocation of the HSM into emergency offline containment.

* **EVI 010:**
    * **Source Type:** Live System Alerts
    * **Timestamp:** 2026-08-26T23:15:00Z
    * **Description:** Security alert reporting that access logs were purged from 19:00 to 22:00 via an SQL injection attack launched from an internal security terminal.
    * **Status:** Altered
    * **Reliability Score:** 0.45
    * **Cross Reference:** EVI 012
    * **Anomaly Note:** Database wipe attempt executed using administrator rights belonging to Kaelen Voss.

* **EVI 011:**
    * **Source Type:** Witness Statement
    * **Timestamp:** 2026-08-26T20:10:00Z
    * **Description:** A junior technician states seeing Marcus Vance hand an encrypted USB flash drive to an unidentified individual near elevator shaft 3.
    * **Status:** Genuine
    * **Reliability Score:** 0.55
    * **Cross Reference:** EVI 007
    * **Anomaly Note:** Indicates physical collusion and unauthorized tool deployment by Vance.

* **EVI 012:**
    * **Source Type:** Device Telemetry
    * **Timestamp:** 2026-08-26T23:10:00Z
    * **Description:** Database query audit log showing user "kvoss" executed a manual update script targeting timestamp indices for RFID reader 7B.
    * **Status:** Genuine
    * **Reliability Score:** 0.95
    * **Cross Reference:** EVI 001
    * **Anomaly Note:** Proves Kaelen Voss directly tampered with RFID logs to fabricate Thorne's entry record.

* **EVI 013:**
    * **Source Type:** Live System Alerts
    * **Timestamp:** 2026-08-26T22:47:00Z
    * **Description:** System alert reporting HSM 09-Alpha heartbeat loss detected on the primary rack position.
    * **Status:** Genuine
    * **Reliability Score:** 0.45
    * **Cross Reference:** EVI 009
    * **Anomaly Note:** Triggers the original automated theft alert despite the unit being moved to the offline vault.

* **EVI 014:**
    * **Source Type:** CCTV Feed
    * **Timestamp:** 2026-08-26T19:48:00Z
    * **Description:** Camera CAM-12 (Server Room B) shows Kaelen Voss logged into the administrative console during his claimed maintenance window.
    * **Status:** Genuine
    * **Reliability Score:** 0.85
    * **Cross Reference:** EVI 012
    * **Anomaly Note:** Verifies Voss's physical presence at the control console during database record manipulation.

* **EVI 015:**
    * **Source Type:** Anonymous Tip
    * **Timestamp:** 2026-08-26T22:00:00Z
    * **Description:** Encrypted email to the facility director claiming Elena Rostova was seen carrying a heavy metallic briefcase out to the main parking lot.
    * **Status:** Completely Misleading
    * **Reliability Score:** 0.30
    * **Cross Reference:** EVI 009
    * **Anomaly Note:** Disproved by B-4 vault telemetry showing the HSM unit remained secured inside sub-basement 4.

TIMELINE CONTRADICTION MATRIX

* **Impossible Gap 1:**
    * **Timeframe:** 19:54 to 19:57
    * **Event A:** Badge EMP-8842 scanned inside Sub-level 2, Vault 9-Alpha.
    * **Event B:** Badge EMP-8842 scanned at the Ground Floor Cafeteria terminal.
    * **Travel Time Required:** Minimum 12 minutes through 3 biometrically gated airlocks.
    * **Resolution:** The Cafeteria RFID scanner was spoofed or cloned; the log entry was inserted by Kaelen Voss to frame Dr. Thorne.

* **Impossible Gap 2:**
    * **Timeframe:** 19:42 to 19:42
    * **Event A:** Badge EMP-8842 logged at the Sector 7 turnstile.
    * **Event B:** Dr. Thorne's mobile device logged on Metro Tower 4, 3.2 miles away.
    * **Travel Time Required:** Minimum 15 minutes transit time.
    * **Resolution:** Badge EMP-8842 was cloned; Dr. Thorne was not present on campus.

VERDICT OPTIONS

* **Option A:**
    * **Claim:** Dr. Aris Thorne physically stole the HSM to sell cryptographic keys.
    * **Validity:** False

    * **Flaw:** Disproved by cell tower telemetry, gait analysis mismatch, and tampered database logs.

* **Option B:**
    * **Claim:** Marcus Vance manipulated the security database and directly stole the HSM.
    * **Validity:** False
    * **Flaw:** Vance provided network access via a compromised hardware tap, but did not move or extract the physical HSM.

* **Option C:**
    * **Claim:** Elena Rostova executed an unauthorized emergency extraction of the HSM to prevent exfiltration via a zero-day exploit, enabled by Marcus Vance's compromised network tap. Meanwhile, Kaelen Voss tampered with database logs and badge data to frame Dr. Thorne and obscure the breach.
    * **Validity:** True
    * **Evidence Chain:** EVI 004, EVI 006, EVI 007, EVI 009, EVI 012
    * **Calculated Confidence Score:** 89.2%

* **Option D:**
    * **Claim:** Kaelen Voss stole the HSM while remote-wiping the security database.
    * **Validity:** False
    * **Flaw:** Voss manipulated database logs to assist the cover-up and frame Thorne, but never physically accessed Vault 9-Alpha or handled the HSM unit.